Programming For Data Science

Part 2 : Flight Data Analysis (2004 ~ 2008)

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Question 2a

In this question, we analyze flight data from 2004 to 2008 to identify the best days and times to minimize delays each year. Before diving into the analysis, we need to establish benchmarks to define what constitutes a short or long delay due to its subjectivity. According to the U.S. Department of Transportation (DOT), flights delayed by 15 minutes or less are considered "on-time," while the Flight Aviation Administration (FAA) treats delays of 45 minutes or more as operationally significant. Based on these standards, we categorize delays and focus on the arrival delays, which reflects the total delay experienced from departure to the arrival of the flight.

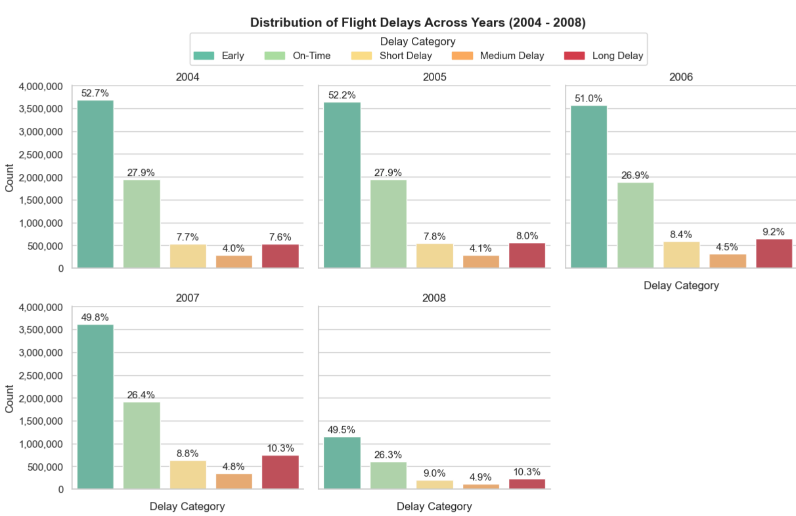
Pre-Processing Steps

Here we proceed to prepare the data by filtering it and adding required columns for analysis to allow for more accurate and better analysis. The pre-processing steps are as follows :

1. **Removing unnecessary columns not ussd for analysis such as TaxiIn, TaxiOut, CancellationCode, and keeping only the arrival delay of all the delay types.**
2. **Filtering data to exclude the flights that are cancelled and diverted (for now).**
3. **Filtering again to remove rows with departure and arrival time values of at least 2400 which does not make sense since a day ends at 2359.**
4. **Categorising the arrival delays and arranging them in order where negative values are considered as early, "On-Time" as within 15 mins, short delays as between 16 and 29 minutes , medium delays as between 30 and 44 minutes, and lastly long delays as more than 45 minutes.**
5. **Converting the year and delay category columns to factors for plotting of graphs.**

Exploratory Data Analysis of Delay Categories

To better understand the current distribution of flight delays, we construct a faceted bar graph showing the number of flights for each delay category every year as shown below. From the diagram, we can see that most of the flights have been early throughout the years, taking up about 50% of the total flights each year. While the early and "On-time" categories have been consistently the highest of the delay categories, it is important to note that the number of **flights experiencing long delays are generally increasing relative to the total number of flights each year** as shown in the increase of its percentage over the years from 7.6% in 2004 to 10.3% in 2008, having the highest percentage amongst the short, medium and long delay categories as of 2008. As such, we aim to find the day of week and time of day to reduce the number of flights experiencing long delays as highlighted from the bar graphs.

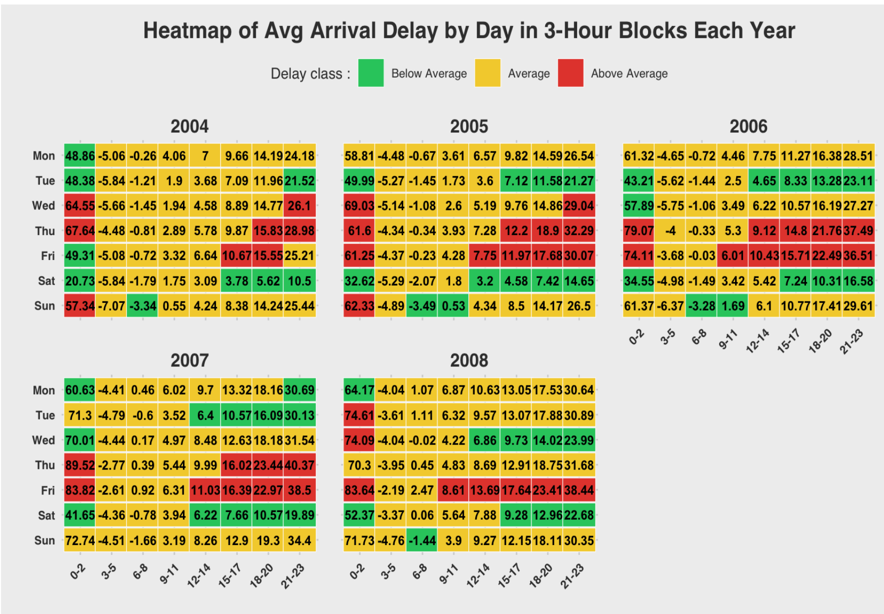


Preparing Data For Time Analysis

To find the best day of the week and time of day to minimise delays, we would require the departure and arrival hours of the different days of the week. Therefore, we extract the departure and arrival hours from the DepTime and ArrTIme columns in the filtered dataframe and group them into 3-hour bins (e.g. 3 - 5), including hours of both ends, for a cleaner analysis. We also renamed and ordered the values in the DayofWeek column to their respective days from Monday to Sunday for better readability. Additionally, we also calculate the average arrival delays (avg_arr_delay) for plotting using the summarise function by grouping the arrival delay variable by the year, day of week and the 3-hour bins (hr_grp), and further grouping the avg_arr_delay again to calculate the average of the arrival delays within each hr_grp for better analysis of the mean for each hour within every hr_grp and call it block_average_delay.

Analysis of Average Arrival Delays Across Hours & Days of Week

Lastly, we then prepare for the plotting by creating a new categorical variable, delay_class, using the block_average_delay and avg_arr_delay values where we identify the bins with higher and lower than average arrival delays across the hr_grp of each year. We define the average category as within 2 minutes difference between the avg_arr_delay and the block_avg_delay values and anything more and less than 2 minutes to be categorised as above and below average respectively. We use these categories to then colour the bins according to the respective delay_class and hence better identify the hours of day and day of week to minimise arrival delays. Therefore, this creates a heatmap for each year as shown below. The green bins represent below average delays, yellow for average delays and red for above average delays. From the visual, it shows that the **0 to 2 hour group, from 0000 to 0259, has the highest number of days where the flights experience above average delays for the hour group as compared to the other hour groups across the years**, highlighting its significance. Instead, **travellers can fly during the next 2 3-hour groups from 0300 all the way till 0859 where flights arrive mostly earlier than expected, if not, experience minor delays.** Therefore, a change in the allocation of flight timings would help reduce the average arrival delays greatly.



In terms of the days, **Thursday and Friday have the most number of hour groups that experience higher than average delays across the years.** Alternatively, **Tuesday and Saturday** consistently showed **lower average delays** and hence travelliers should chooe to travel during those days. A further analysis of the block average delay and the overall block average delay for each year complements the heatmap above as shown below as the colors of the heatmap are based on the "Block Avg Delay" column of each year. From the values, we can hence conclude that the overall block average delay is generally increasing from 12.3 mins in 2004 to 18 mins in 2008. This could be due to the fact that the first 3-hour block has the highest block average across the different hour groups of each year, increasing from 52 mins in 2004 to 71 mins in 2008. In conclusion, best times and days to travel to minimise delays would hence be on **Tuesday and Saturday from 0300 to 0859 or 1500 to 2359 on the same days** since they appear to experience early flights on average and seem to be consistently below the average block average delay for the hour groups.

```
===== Year: 2004 =====
# A tibble: 8 × 3
  hr_grp `Block Avg Delay` block_delay_category
  <fct>              <dbl> <fct>
1 0-2                 52   Long Delay
2 3-5                 -5.5 Early
3 6-8                 -1.3 Early
4 9-11                 2.4 On-Time
5 12-14                5   On-Time
6 15-17                8.4 On-Time
7 18-20               13.4 On-Time
8 21-23               23.8 Short Delay
```

```
===== Year: 2005 =====
# A tibble: 8 × 3
  hr_grp `Block Avg Delay` block_delay_category
  <fct>              <dbl> <fct>
1 0-2                 57.3 Long Delay
2 3-5                 -4.8 Early
3 6-8                 -1.3 Early
4 9-11                 2.7 On-Time
5 12-14                5.4 On-Time
6 15-17                9.2 On-Time
7 18-20               14.4 On-Time
8 21-23               26.4 Short Delay
```

```
===== Year: 2006 =====
# A tibble: 8 × 3
  hr_grp `Block Avg Delay` block_delay_category
  <fct>              <dbl> <fct>
1 0-2                 60.8 Long Delay
2 3-5                 -5   Early
3 6-8                 -1.1 Early
4 9-11                 3.9 On-Time
5 12-14                7.1 On-Time
6 15-17               11.3 On-Time
7 18-20               17.1 Short Delay
8 21-23               29.2 Short Delay
```

```
===== Year: 2007 =====
# A tibble: 8 × 3
  hr_grp `Block Avg Delay` block_delay_category
  <fct>              <dbl> <fct>
1 0-2                 72.1 Long Delay
2 3-5                 -4   Early
3 6-8                 -0.1 Early
4 9-11                 4.8 On-Time
5 12-14                8.6 On-Time
6 15-17               12.9 On-Time
7 18-20               18.7 Short Delay
8 21-23               33   Medium Delay
```

```
===== Year: 2008 =====
# A tibble: 8 × 3
  hr_grp `Block Avg Delay` block_delay_category
  <fct>              <dbl> <fct>
1 0-2                 71   Long Delay
2 3-5                 -3.7 Early
3 6-8                  0.6 On-Time
4 9-11                 5.8 On-Time
5 12-14                9.5 On-Time
6 15-17               12.6 On-Time
7 18-20               17.7 Short Delay
8 21-23               30.3 Medium Delay
```

```
  Year  overall_delay
  <fct>         <dbl>
1 2004           12.3
2 2005           13.7
3 2006           15.4
4 2007           18.3
5 2008           18
```

Question 2b

We are now tasked to evaluate whether the older planes suffer more delays from a year-to-year basis. Similar to the previous part, since age of a plane is subjective, we refer to Paramount Business Jets, stating that a new plane is within 10 years of age, a standard plane is within 11 to 19 years and an old plane is 20 years or older.

Pre-Processing Steps

Before we begin to analyse the plane's data, we first need to prepare the data again as we require the use of another dataset. As such, here are the pre-processing steps made in the planes_df dataframe :

1. **Removing rows with empty and null values. After exploring, we realise that there are 549 rows of data filled with just spaces. After trimming the spaces, we categorise the "None" and "0000" values of the "Year" column as NA and remove them as a whole. At the same time, we also remove the NA and "None" values from the issue date column.**
2. **Creating a new column, flight_date, in the flights_filtered dataframe from part a, combining the values from the Year, "DayOfMonth" and Month columns in the mm-dd-yyyy format. We also do the same for the issue data column in the planes_df dataframe, converting the issue date column in the same format as the flight_date column.**
3. **Renaming the tailnum column to "TailNum" for joining with the flights_filtered dataframe later on.**
4. **Filtering NA values from the flights_filtered dataframe after the joining process as some planes do not have an issue date attached to them in the dataframe**
5. **Categorising the flights_age by New, Mid-age & Old planes according to the values as shown above by Paramount Business Jets, where a new plane is within 10 years of age, a standard plane is within 11 to 19 years and an old plane is 20 years or older.**
6. **We also removed some planes that have issue dates later than when they were used, leading to a negative age as we used the last_flight_date of each year to see how old the planes are.**

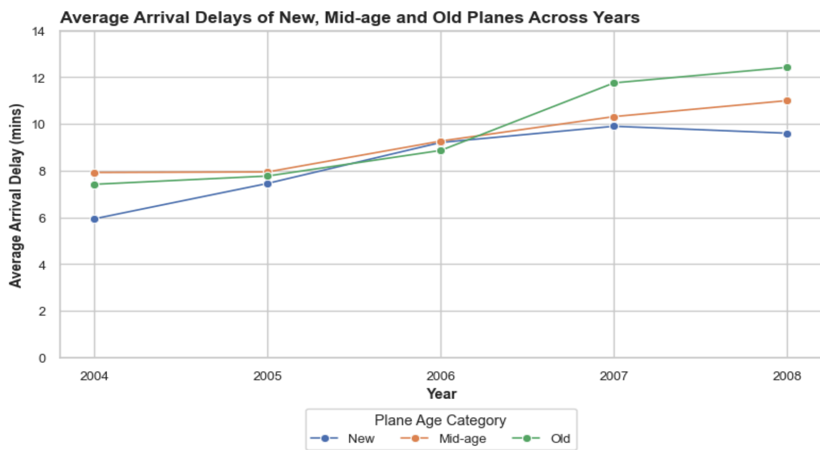
Exploratory Data Analysis of distribution of plane age categories

As seen from the figure below, we explore the number of planes of each age category across the years. We use the latest flight date of each TailNum to calculate the flight age of the planes of each year. We do this by using the latest flight date and the issue date of the TailNum to count the difference of days between the 2 dates and divide it by 365.25 to count the number of years the plane has aged and round them off to 2d.p. , accounting for the leap years.However, this would mean that if the plane did not fly during the year, it would not be counted as part of the count of each year. From the diagram, most of the planes are found to be new with a general increase from 2138 in 2004 to 3188 in 2008. The old planes have the lowest count as compared to the other plane age categories throughout the 5 years. As we observe an increase in the total number of planes each year, this could mean that there was an increasing number of flights over the years due to the nature of the calculation of the flight_age of the planes using their last flight dates.

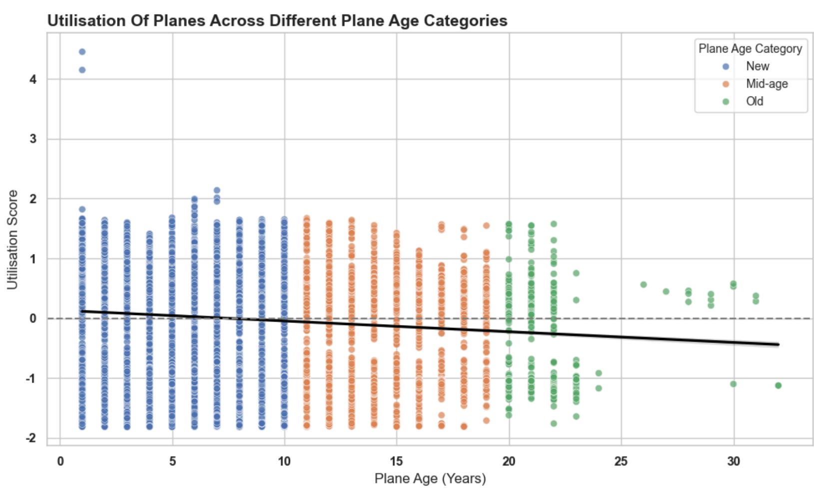
	Year	flight_age_category	total_planes
	<fct>	<fct>	<int>
1	2004	New	2138
2	2004	Mid-age	676
3	2004	Old	5
4	2005	New	2357
5	2005	Mid-age	723
6	2005	Old	38
7	2006	New	2608
8	2006	Mid-age	760
9	2006	Old	104
10	2007	New	3150
11	2007	Mid-age	889
12	2007	Old	163
13	2008	New	3188
14	2008	Mid-age	928
15	2008	Old	180

Analysis of Flight Arrival delays Across Plane Age Categories

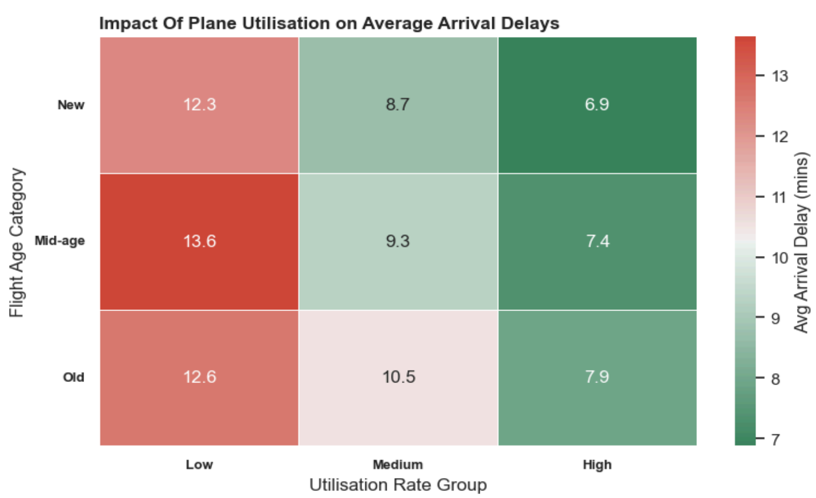
Using the plane age categories, arrival delay and year, we calculate the average arrival delay of each age category across the years. This will thus give us the line plot as shown below where we plot the year against the average arrival delay across the different plane age caategories. As seen from the plot, we can observe that all 3 categories face an increasing trend of average arrival delays with the old planes suffering the longest average delays out of the 3 by the end of 2008. While the old and mid-age planes had an increase in their average arrival delays from 2007 to 2008, the new planes on the other hand had a slight improvement in terms of average arrival delays, showing that the new planes could have a slight advantage due to their young age as compared to the other plane age categories. In 2006, there was a huge increase in the average arrival delays of the old planes, beating the new and mid-age plane categories from being the lowest out of the 3. This proves the significance of the increase in average arrival delays for the old planes. By 2008, the plane age categories are organised in ascending order based on the average arrival delays, with the younger planes experiencing the least delays and the older planes facing progressively longer average arrival delays. As such, to a certain degree, we can say that the older planes do suffer from more delays.



However, aircraft age alone may not fully capture the operational condition of a plane, as it is also influenced by utilisation. In particular, aircraft that are used more intensively may experience greater wear and tear, and could therefore be considered “older” in a functional sense despite having a lower chronological age.



To assess whether utilisation contributes to the delays observed, the relationship between aircraft age and utilisation was examined using the scatter plot as shown. From the plot, we can observe a weak negative correlation (-0.11) between the plane age (years) and their utilisation scores, which is calculated by equally weighing the standardised the number of flights of each tailnum and their distance covered, suggesting that older aircraft are not more heavily utilised and may, in fact, be slightly less utilised than newer ones. This implies that the higher delays experienced by older aircraft are unlikely to be driven by utilisation.



To further discuss this, we also plot a heatmap of the plane utilisation categories against the different plane age categories and the average arrival delys of the flights, revealing that higher utilisation is mostly associated with lower average delays across all age categories and vice versa, except for the low utilisation group where the mid-age planes (11 to 19 years) experiences the greatest average arrival delay (13.6 mins) across all utilisation groups and plane age categories. This suggests that highly utilised aircraft could be more operationally efficient or better maintained, as they are likely deployed on more optimised schedules and subject to stricter maintenance regimes. Consequently, the increased delays observed in older aircraft are more likely to be attributable to age-related factors such as mechanical wear, increased maintenance requirements, and reduced reliability, rather than utilisation intensity.

	sum_sq	df	F	PR(>F)
C(plane_age_category)	3868.849322	2.0	34.142867	1.592637e-15
Residual	947981.115927	16732.0	NaN	NaN

Multiple Comparison of Means - Tukey HSD, FWER=0.05

group1	group2	meandiff	p-adj	lower	upper	reject
Mid-age	New	-0.9126	0.0	-1.2498	-0.5754	True
Mid-age	Old	1.0186	0.0092	0.2067	1.8306	True
New	Old	1.9312	0.0	1.1601	2.7023	True

To finally conclude the evaluation if older planes suffer from more delays on a year to year basis, a one-way ANOVA was conducted to compare average delays across plane age categories. The results show a statistically significant difference in delays ($F = 34.14$, $p < 0.001$). The F-statistic indicates that the variation in delays between age groups is substantially larger than the variation within each group, suggesting that the differences are meaningful rather than due to random fluctuations. The very small p-value further confirms that there is less than a 0.1% probability that these differences occurred by chance.

However, while ANOVA confirms that differences exist, it does not indicate which specific groups differ. Therefore, a Tukey HSD post-hoc test was conducted to perform pairwise comparisons between age categories while controlling for multiple comparisons. The results show that all pairwise differences are statistically significant. In particular, older aircraft experience significantly higher delays than both mid-age and newer aircraft, with the largest difference observed between new and old aircraft (approximately 1.93 minutes). Therefore, these findings reinforce the conclusion that delays increase with aircraft age.

Question 2c

Assuming that we are predicting the probability of diversions before departure, we carefully select the necessary columns for the logistic regression model to prevent any data leakages that would not be available at the time of prediction such as the different forms of delay and the actual departure and arrival times. As such, these are the features selected for the model as shown below with new derived features to enhance the predictive capability of the model such as route and the destination arrival hour, which will be further explained later.

- **Temporal factors :** scheduled departure and arrival hours, Destination arrival hour, Origin departure hour, Month
- **Operational factors :** carrier
- **Network / Geographic factors :** route, Origin, Destination, Distance
- **Historical Performance factors :** historical route and destination diversion rates

Before we head on to analysing the model, we first analyse the distribution of actual diverted flights across the years to help identify patterns, class imbalances and ensure that the model is built on stable and meaningful data as shown below.

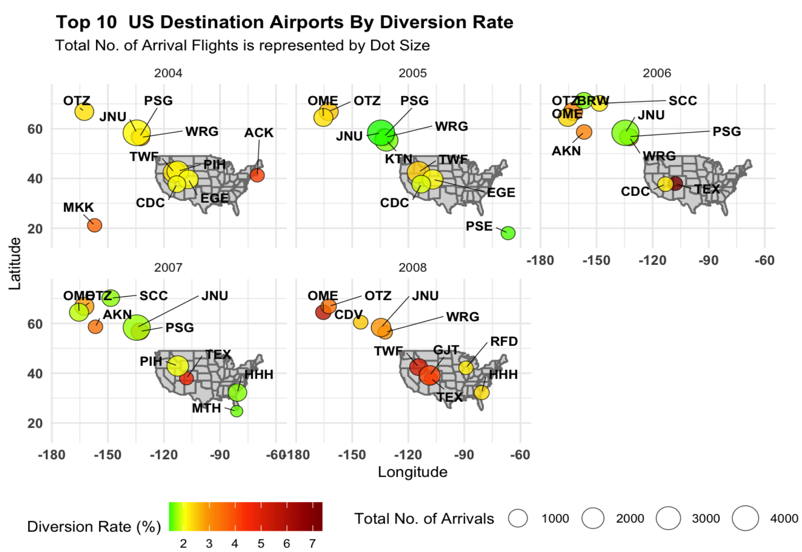
	Year	total_flights	div_count	div_percent
0	2004	6993604	13784	0.197094
1	2005	6996456	14028	0.200502
2	2006	7013271	16186	0.230791
3	2007	7289022	17179	0.235683
4	2008	2323620	5654	0.243327

From the image, we can observe that the diversion rate, div_percent, generally ranges from 0.197% in 2004 to 0.243% in 2008, showing a clear upward trend of diversion across the years, although the diversion rates remain relatively low. However, while there was an increase in the diversion rate of the flights, there seems to be a sharp fall in the number of flights from ~7.3 million flights in 2007 to ~2.3 million flights in 2008. As such, the increase in the diversion rate in 2008 should be treated with caution since the lower total flight count increases the sensitivity of the diversion rate to fluctuations, meaning that the relatively small changes in the number of diverted flights can result in disproportionately large increases in the overall percentage of the diverted flights.

After getting a better understanding of the diversion rates, we now explore the different features to better understand their relationships and impact on the diversion rates. The exploratory analysis also offers early indications of which features may be most informative and relevant for the model.

Destination

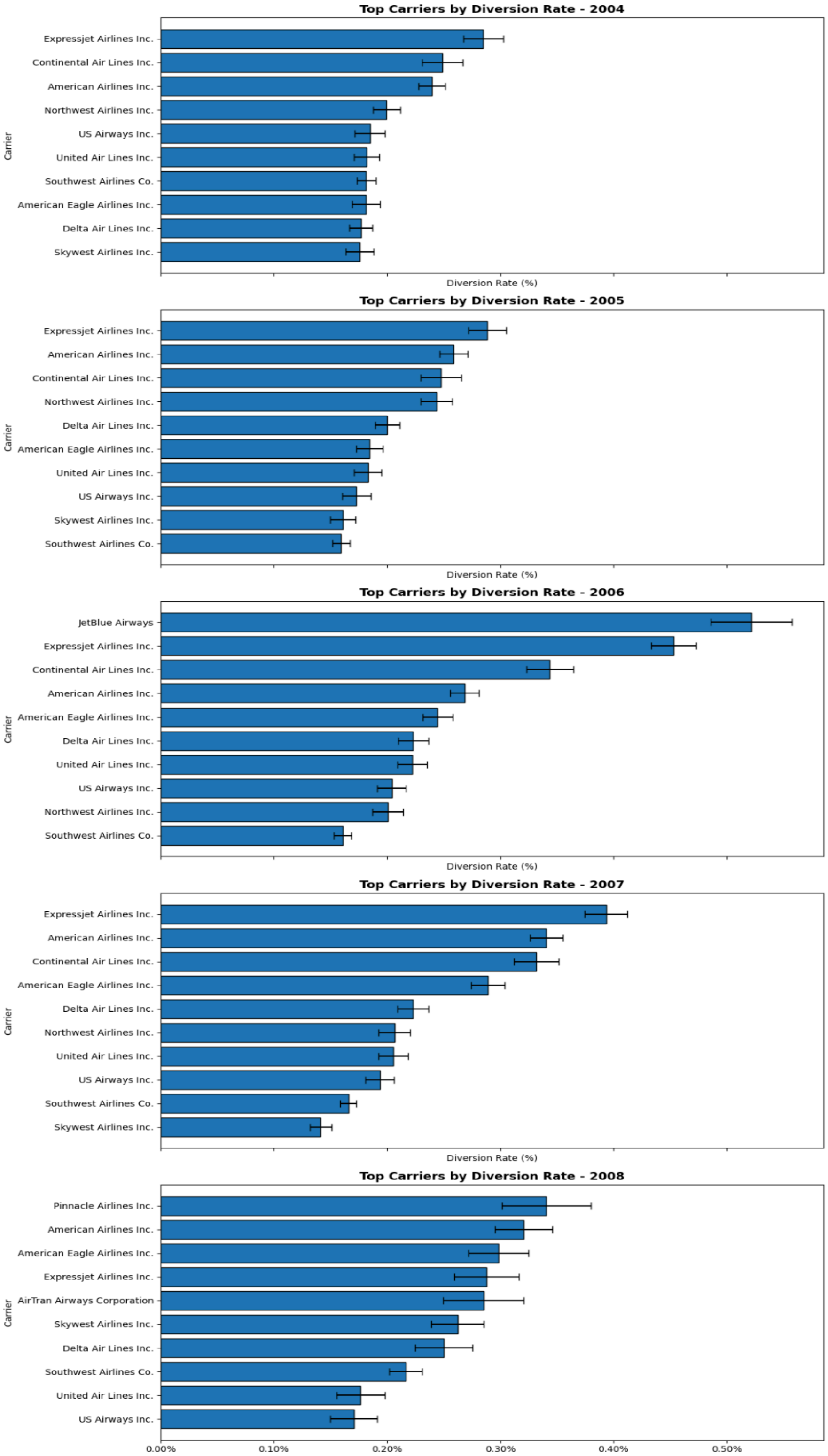
Here we perform exploratory data analysis of the destination airports by their diversion rates. For a cleaner analysis, we reduce the number of unique values of each feature to just the top 10 destination airports in terms of the diversion rates. We also further filter the destination airports to have at least 50 arrivals at each airport since some airports have high diversion rates due to their extremely low number of arrival flights. From the figure on the left, the size of the dot represents the total number of arrival flights of each airport. As we can see from the diagram, the airports on the top



left corner of the map appear consistently as the top ten airports with high diversion rates, such as OTZ, PSG and JHU. While having a high number of arrival flights of at least 1000, this could explain its significance as compared to other airports. In 2006, TEX experienced the highest diversion rate of all time as seen from its dark red colour. Over the years, the diversion rates tend to be more popular around the north-west and central regions of the country with many constant appearances on the map.

To capture these underlying patterns, a feature representing the historical diversion rate at each destination, computed from previous years, was introduced, with the first year using the training data to input into the feature. This engineered feature allows the model to incorporate past performance as a proxy for latent risk factors that are not directly observable in the raw data. By doing so, the model is better equipped to identify airports with inherently higher diversion risk, thereby improving its predictive capability.

Carrier



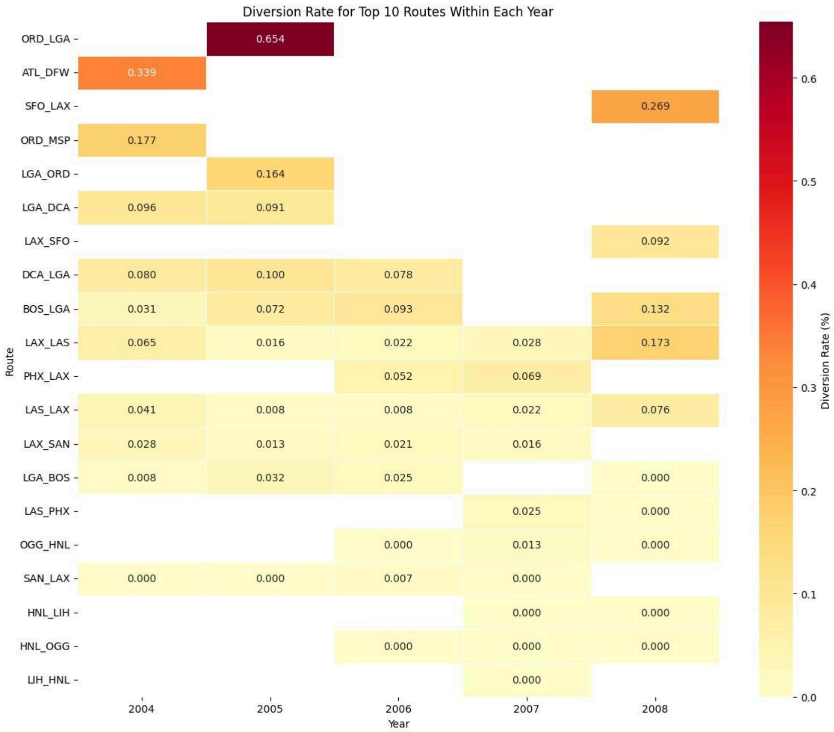
This visual describes the top 10 carriers across the years in descending order of the diversion rates, also after filtering for the carriers with at least 50 flights for each year to prevent outliers with extremely high diversion rates. The distribution of diversion rates across carriers reveals clear differences in relative diversion risk over time. Across multiple years, carriers such as ExpressJet Airlines consistently rank among those with higher diversion rates, with values generally in the range of approximately 0.30% to 0.45%, indicating a relatively higher likelihood of diversion compared to other carriers. In contrast, carriers such as Southwest Airlines and SkyWest Airlines tend to exhibit lower diversion rates, approximately about 0.15% to 0.22%, suggesting comparatively more stable operational performance.

The variation in diversion rates across carriers is evident from the spread of values within each year. For instance, in 2006, JetBlue Airways recorded a diversion rate exceeding 0.50%, while several other carriers in the same year remained closer to the 0.15% to 0.25% range, indicating substantial differences in diversion risk across airlines. Similarly, changes in diversion rates for the same carrier across years, such as ExpressJet Airlines, demonstrated that diversion behaviour is not constant over time. These patterns suggest that diversion risk varies both across carriers and across years, likely influenced by differences in operational factors and changing external conditions.

The error bars represent 95% confidence intervals for the estimated diversion rates, providing a measure of uncertainty around the sample-based estimates of each carrier's true diversion probability. Carriers such as ExpressJet Airlines and American Airlines generally display relatively narrow confidence intervals, indicating that their diversion rate estimates are stable and based on sufficient data. In contrast, carriers such as Pinnacle Airlines and JetBlue Airways exhibit wider confidence intervals, particularly in years where their diversion rates are elevated, suggesting greater variability and less precision in these estimates. This also implies that the observed values may be more sensitive to fluctuations and should be interpreted with greater caution.

Overall, the visual demonstrates that while some carriers consistently exhibit higher diversion rates, the magnitude and statistical certainty of these differences vary across years. The use of confidence intervals ensures that these comparisons account for uncertainty, allowing for more robust and reliable interpretation of carrier-level diversion risk.

Route



The heatmap illustrates the diversion rates of the top 10 routes within each year, where the routes are identified separately for each year based on their diversion rates. We also filter the routes to only routes with a minimum of 50 flights to ensure that there are no extreme outliers due to the low number of flights.

Each row represents a route and each column represents a year, with colour intensity indicating the magnitude of the diversion rate. Cells are left blank where a route does not appear among the top 10 riskiest routes in that particular year, allowing for clearer focus on the most operationally significant routes while still enabling cross-year comparison for routes that persist over time.

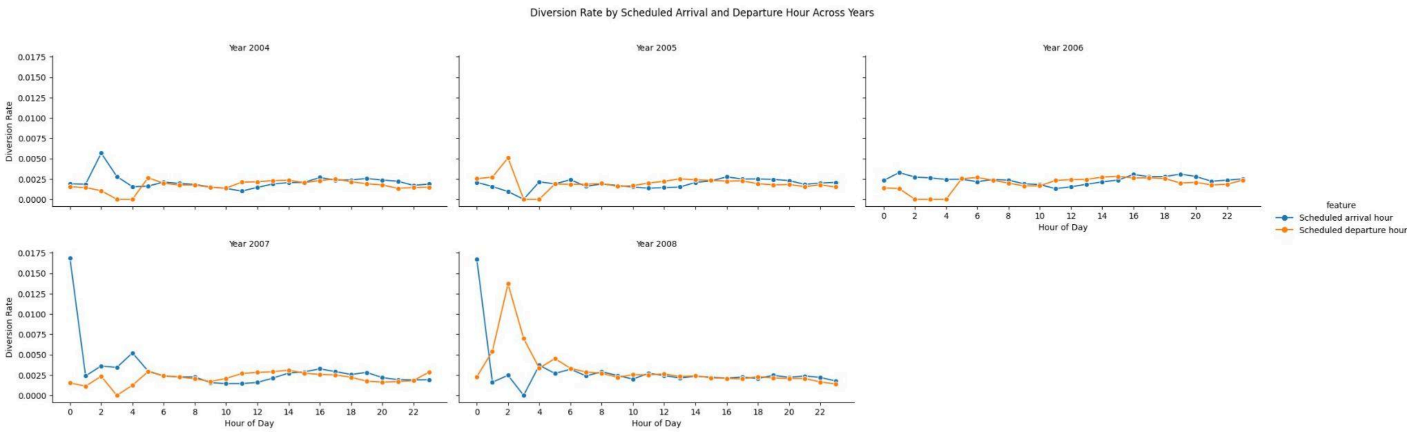
The variation in diversion rates across routes and years is evident from the wide range of observed values, from 0.00% to above 0.65%, indicating substantial heterogeneity in route-level diversion risk. For example, the ORD–LGA route records a notably high diversion rate of approximately 0.654% in 2005, which is more than six times higher than consistently low-risk routes such as LAX–SAN, which remain below 0.05% across most years. Similarly, ATL–DFW (~0.339% in 2004) and SFO–LAX (~0.269% in 2008) demonstrate periods of elevated diversion risk, highlighting the presence of route-specific spikes.

The observed patterns can be broadly categorised into three types. First, consistently low-risk routes, such as LAX–SAN and SAN–LAX, maintain very low diversion rates across all years, suggesting stable operational conditions. Second, gradually evolving routes, such as BOS–LGA and LAX–LAS, show increasing diversion rates over time (e.g. BOS–LGA rising from ~0.031% in 2004 to ~0.132% in 2008), indicating changing operational or environmental conditions. Third, spike-driven routes, such as ORD–LGA and SFO–LAX, exhibit sharp increases in specific years, suggesting sensitivity to episodic disruptions such as weather or congestion.

A route may therefore appear in some years but not others depending on whether it ranks among the highest diversion rates in that particular year. This indicates that the relative risk ranking of routes is dynamic over time, with certain routes experiencing periods of elevated diversion risk that place them among the top routes, while in other years they fall below this threshold. This further reinforces that diversion risk is context-dependent, as routes may move in and out of the highest-risk group across different years. As such, route-based diversion behaviour is not static, and both the level of risk and the relative prominence of a route in terms of diversion risk may shift across years.

Based on these observations, a historical route feature is constructed by aggregating diversion rates from previous years for each route. The persistence observed in certain routes, particularly those that consistently exhibit low or elevated diversion rates, supports the use of historical aggregation as it captures underlying structural tendencies rather than relying solely on short-term fluctuations. At the same time, the presence of variability and spikes indicates that this feature reflects probabilistic tendencies rather than deterministic outcomes. Incorporating this historical information enables the model to leverage both long-term patterns and temporal variation in route behaviour, thereby improving its ability to predict diversion risk.

Scheduled Departure and Arrival hours

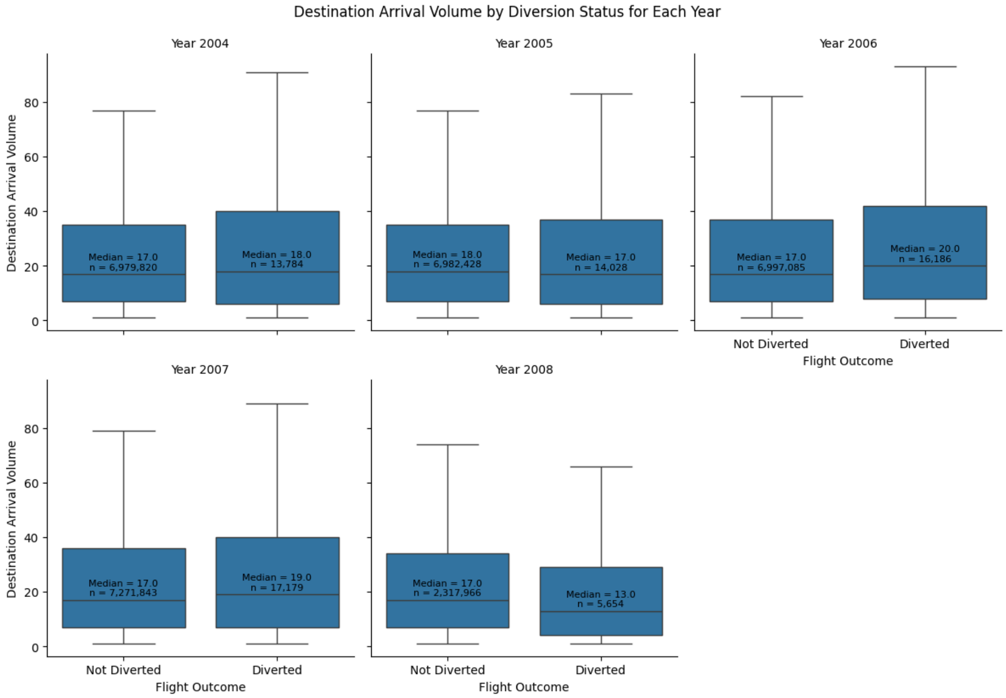


The scheduled departure and arrival hours are derived by extracting the hours from the scheduled departure and arrival times respectively. During data preprocessing, any instances of hour “24” in the scheduled time variables were recorded as hour “0” to maintain a consistent 24-hour time format. This ensures accurate representation and interpretation of time-of-day patterns in the analysis.The plot illustrates the variation in diversion rates across different hours of the day for both scheduled arrival and departure hours over multiple years. Overall, diversion rates remain relatively low throughout the day, generally ranging between approximately 0.10% and 0.30% for most hours. However, distinct temporal patterns and occasional spikes are observed, particularly during early morning hours.

A consistent observation across multiple years is the presence of higher volatility during the early hours of the day - hour 0 to hour 4. For example, in 2007 and 2008, the diversion rates showed sharp spikes exceeding 1.5% – 1.7% at hour 0, which is significantly higher than the typical rates observed during the rest of the day. Similarly, the scehduled departure hour in 2008 exhibits a pronounced spike of approximately 1.3%–1.4% around hour 2, suggesting that disruptions during early operations may have a disproportionate impact on diversion risk.These extreme spikes may be partially influenced by lower flight volumes during early hours, which can lead to greater variability in diversion rate estimates compared to more densely populated mid-day periods. Beyond these early-hour anomalies, diversion rates stabilise considerably throughout the mid-day period - hour 6 to hour 18, where both arrival and departure patterns converge and remain relatively consistent, typically within a narrow band of 0.15% to 0.30%. This suggests that operational conditions during standard flight hours are more stable and predictable.In the later hours of the day (evening to night), there is a slight increase in diversion rates in some years, particularly for arrival hours, which may be attributed to accumulated delays, congestion, or cascading operational disruptions throughout the day. However, these increases are relatively moderate compared to the early-morning spikes.

Overall, the results indicate that diversion risk is influenced by time-of-day effects, with early morning hours showing higher variability and mid-day periods demonstrating more stable behaviour. These patterns suggest that time-based features, such as scheduled departure and arrival hours, contain useful predictive information. While the overall relationship appears relatively stable across most hours, the presence of non-uniform patterns and localized spikes indicates that time-of-day effects are not strictly linear, and may interact with other operational factors in influencing diversion risk.Accordingly, the exploratory analysis focuses on scheduled departure and arrival hours to capture these general time-of-day effects, as they provide a clear and interpretable view of temporal patterns such as early-hour variability and mid-day stability. In addition, interaction features combining the airports and time (e.g., origin–departure hour and destination–arrival hour) are constructed for the model to capture more granular, location-specific temporal effects. While these interaction features are not visualised directly due to their high dimensionality, they enable the model to learn more complex patterns beyond the general trends observed in the exploratory analysis.

Destination Arrival Volume



The boxplots illustrate the distribution of destination arrival volume for diverted and non-diverted flights across each year. Overall, the distributions for both groups are broadly similar, with substantial overlap in their interquartile ranges and whiskers. This suggests that destination arrival volume alone does not strongly distinguish between diverted and non-diverted flights at a general level.

It is also important to note that the number of diverted flights is significantly smaller than the number of non-diverted flights each year. Despite this imbalance, diverted flights in certain years still exhibit higher median arrival volumes compared to non-diverted flights. For instance, in 2006 and 2007, the median arrival volume for diverted flights (20 and 19) exceeded that of non-diverted flights (17 in both years). This suggests that the observed differences are not merely due to sample size imbalance, but may reflect underlying operational factors. One possible explanation is that flights arriving at busier destinations are more susceptible to congestion, capacity constraints, or air traffic control restrictions, which may increase the likelihood of diversion.

Small differences in median values can be observed across years. In 2004, diverted flights had a slightly higher median arrival volume (18 vs 17), while in 2005 the relationship reversed (17 vs 18). In 2006 and 2007, diverted flights again showed higher median values, suggesting that flights arriving at busier destinations may be more prone to diversion in these years. However, in 2008, diverted flights exhibit a lower median arrival volume (13 vs 17), indicating a reversal in this pattern. Despite these fluctuations, the differences in medians are relatively small and the distributions remain highly overlapping. This indicates that while destination arrival volume may have some influence on diversion outcomes, it is unlikely to be a strong standalone predictor and may interact with other operational factors. In addition, the whiskers for diverted flights in certain years appear slightly more extended compared to those for non-diverted flights, indicating greater variability and the presence of more extreme values. This suggests that diversion outcomes may be associated with a wider range of operational conditions, particularly at higher arrival volumes.

While the boxplots provide a visual comparison of the distributions, they do not indicate whether the observed differences are statistically meaningful. Therefore, a Mann–Whitney U test is conducted to formally assess whether the differences between diverted and non-diverted flights are statistically significant, without assuming normality of the data as seen below.

	Year	Median (Not Diverted)	Median (Diverted)	Median Difference	U Statistic	P-Value	Significance
0	2004	17.0	18.0	1.0	48,003,660,928	0.6688	ns
1	2005	18.0	17.0	-1.0	49,618,284,940	0.0071	**
2	2006	17.0	20.0	3.0	54,191,307,760	2.748e-21	***
3	2007	17.0	19.0	2.0	61,314,515,615	3.112e-05	***
4	2008	17.0	13.0	-4.0	7,224,612,414	1.378e-40	***

Statistical significance is represented using conventional thresholds, where *** indicates p-values less than 0.001, ** indicates p-values less than 0.01, * indicates p-values less than 0.05, and “ns” denotes non-significance. The statistical results reveal that in 2004, the difference in median arrival volume between diverted and non-diverted flights is not statistically significant ($p = 0.6688$), indicating no meaningful distinction between the two groups. However, from 2005 onwards, statistically significant differences emerge. In 2005, diverted flights exhibit a slightly lower median arrival volume (17 vs 18, difference = -1), and this difference is statistically significant ($p = 0.0071$). The effect becomes more pronounced in 2006 and 2007, where diverted flights have higher median arrival volumes (20 vs 17 and 19 vs 17, respectively), with highly significant p-values ($p < 0.001$). This suggests that in these years, higher traffic at destination airports may be associated with increased diversion risk. In 2008, the relationship reversed, with diverted flights showing a substantially lower median arrival volume (13 vs 17, difference = -4), again with strong statistical significance ($p < 0.001$). This indicates that the relationship between arrival volume and diversion is not consistent across years.

Despite the statistical significance observed in several years, the magnitude of the differences remains relatively small in most cases. This suggests that while destination arrival volume contributes some predictive information, its practical impact may be limited when considered in isolation.

Construction of Logistic Regression Model and its Analysis

After conducting exploratory data analysis of the features of the model to explore their relationships with the diversion rates, we then produce the logistic regression model that is able to predict the probability of a diversion occurring.

We first begin preparing the data for the creation of the model by selecting only the necessary columns from the div_df dataframe for the features of the model. The features are subsequently categorised into categorical and numerical variables to facilitate appropriate preprocessing. Categorical features are transformed using one-hot encoding, while numerical features are standardised to ensure consistent scaling across variables. In particular, variables such as month and scheduled departure and arrival hours are treated as categorical rather than numerical. Although these variables are represented numerically, they do not exhibit a meaningful linear or ordinal relationship with diversion risk as seen from the exploratory analysis earlier. Instead, each value represents a distinct time category, where patterns may vary non-linearly across different months or hours of the day. Therefore, treating them as categorical variables allows the model to capture these distinct effects without imposing an artificial linear structure.

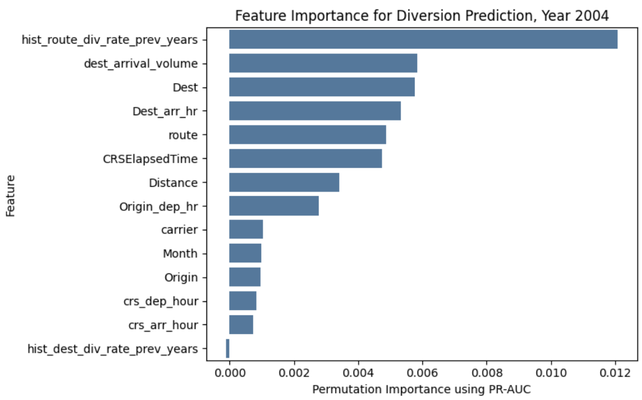
After preprocessing the categorical and numerical variables, yearly logistic regression models were trained to predict whether a flight would be diverted. For each year, the dataset was split into training and testing sets using a 70:30 split, with stratification applied to preserve the proportion of diverted and non-diverted flights in both sets. This is important because flight diversions are rare events, and an unstratified split could result in an uneven distribution of diverted cases. The model was also trained with class weighting to account for the imbalance in the dataset, ensuring that the minority class (diverted flights) is given appropriate importance during training.

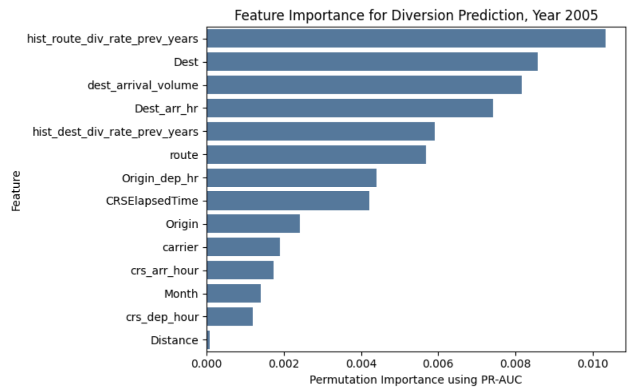
Historical route and destination diversion-rate features were then constructed to incorporate past operational behaviour into the model. For later-year models, the historical diversion rates were calculated using only data from previous years. For example, the 2006 model used historical route and destination diversion patterns from 2004 and 2005 only. This prevents data leakage, as the model does not use information from the year it is trying to predict or from future years. Since 2004 is the first year in the dataset and has no prior-year data, its historical route and destination features were created using only the 2004 training data. A fallback diversion rate was also calculated from the training data to handle routes or destinations that did not appear in the historical records. To reduce instability from low-frequency routes or destinations, smoothing was applied when calculating the historical rates. This prevents routes with very few flights from having disproportionately extreme historical diversion rates. The historical route diversion rate was calculated as a smoothed proportion of past diverted flights over total past flights for each route. Similarly, the historical destination diversion rate was calculated based on past diversion behaviour at each destination airport. These engineered features allow the model to capture route-specific and destination-specific risk patterns observed during the exploratory analysis.

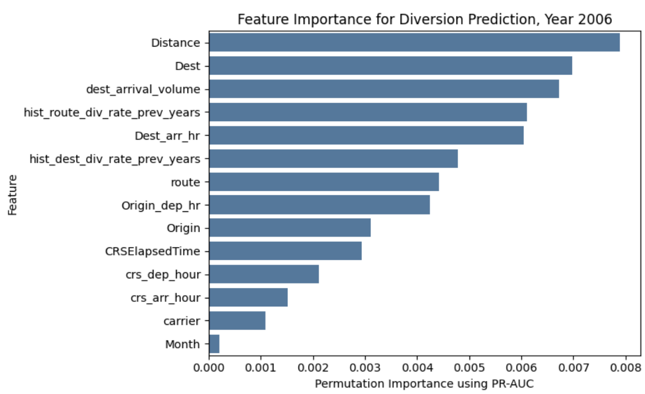
Model performance was evaluated using a range of metrics, including the confusion matrix, precision, recall, F1-score, ROC-AUC, and PR-AUC. Given the class imbalance in the dataset, particular emphasis was placed on precision, recall, and PR-AUC, as these metrics provide a more informative assessment of the model’s ability to correctly identify the minority class (diverted flights). In contrast to ROC-AUC, which can appear overly optimistic in imbalanced settings, PR-AUC offers a more sensitive measure of performance by focusing on the trade-off between precision and recall.

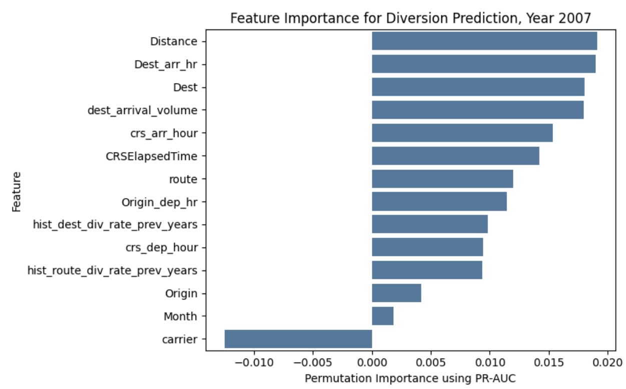
Across all years, the model demonstrates consistently high recall for the minority class, ranging from approximately 0.45 to 0.62, indicating that a substantial proportion of diverted flights are correctly identified. However, this comes at the expense of very low precision, which remains below 0.01 in all years, reflecting a high number of false positives. This trade-off is expected in imbalanced classification settings, where the model prioritises capturing rare events over maintaining precision. Overall model accuracy remains relatively high, ranging from approximately 0.73 to 0.84, although this is largely driven by the dominance of the non-diverted class. The F1-score for the diverted class remains low across all years (approximately 0.008 to 0.014), reflecting the inherent difficulty of predicting rare diversion events, where achieving both high precision and high recall simultaneously is challenging. While the model is able to identify a substantial proportion of diverted flights (moderate recall), the low precision indicates a high number of false positives, resulting in a lower overall F1-score. This highlights the trade-off between capturing rare events and maintaining prediction accuracy.

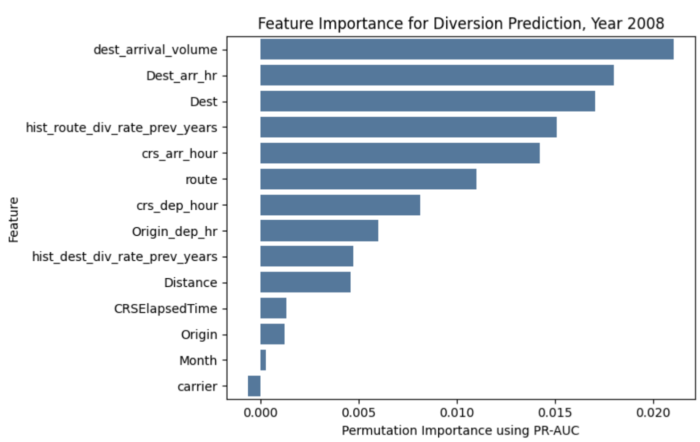
We also plot the feature importance across the years as seen below.











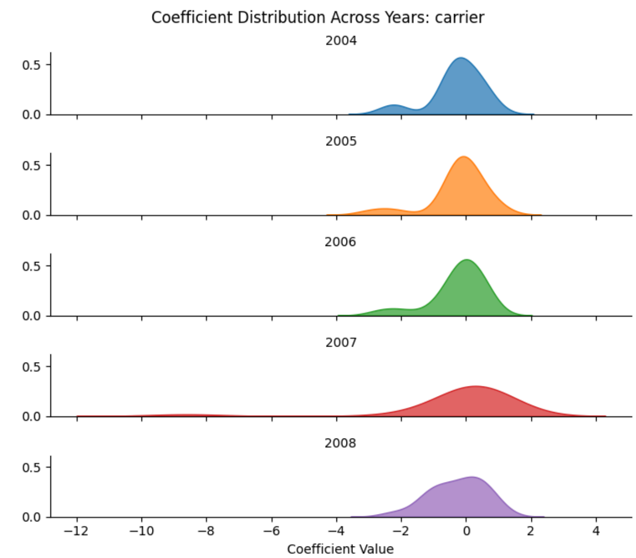
The permutation importance results provide further insight into which features contribute most to the model's predictive performance across different years. Overall, a consistent pattern emerges in which destination-related features, historical features, and operational variables play a dominant role in predicting diversion outcomes.

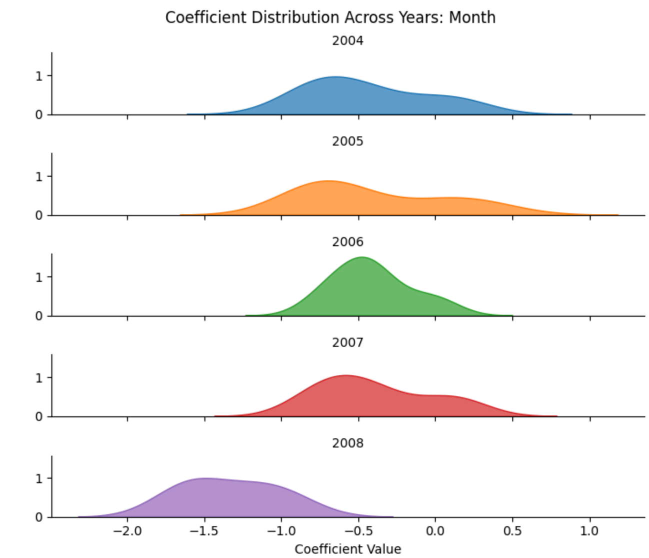
In the earlier years, particularly 2004 and 2005, the historical route diversion rate was the most influential feature (importance ≈ 0.012 in 2004 and 0.010 in 2005). This highlights the strong predictive value of past route-level diversion behaviour, supporting the inclusion of the engineered historical feature. Destination-related variables such as destination airport, destination arrival volume, and destination–arrival hour also rank highly, indicating that location-specific and congestion-related factors are important drivers of diversion risk. In 2006, the importance of features becomes more distributed, with distance emerging as the most influential variable (importance ≈ 0.008), alongside destination-related features and historical features. This suggests that route characteristics, such as flight length, may also play a role in diversion likelihood during certain periods.In 2007 and 2008, destination and temporal interaction features become increasingly dominant. Features such as destination arrival hour, destination airport, and destination arrival volume consistently rank among the top predictors, with importance values exceeding 0.017–0.021. This indicates that time-dependent and location-specific patterns are critical in explaining diversion behaviour in later years.

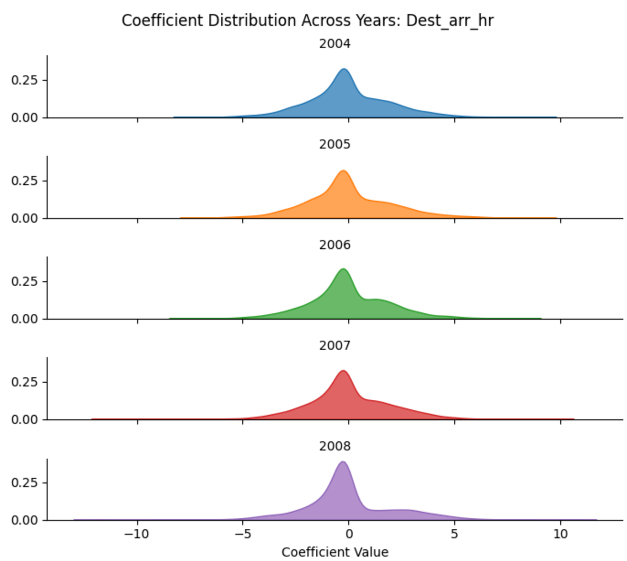
While the historical route and destination diversion rates remain important across all years, their relative importance becomes more moderate in later years as other operational and temporal features gain prominence. This suggests that historical patterns provide a strong baseline signal, but must be considered alongside dynamic factors such as time-of-day and airport conditions. In contrast, features such as month, carrier, and origin consistently show low or negligible importance across all years. In some cases, such as the carrier variable in 2007, negative importance values are observed, indicating that the feature may introduce noise rather than meaningful predictive information when other variables are included.

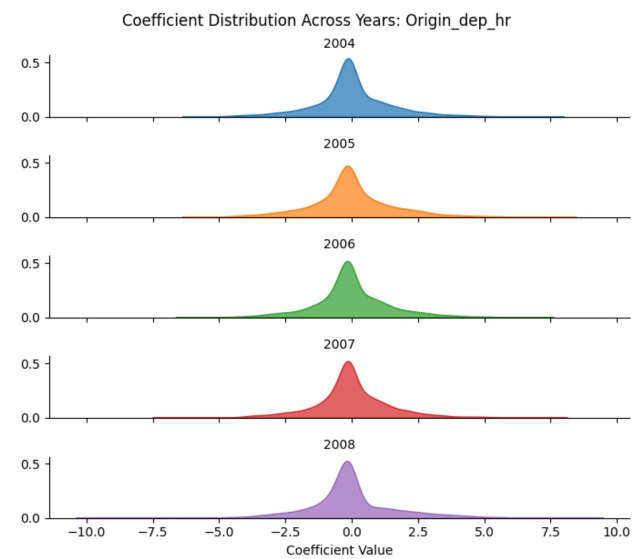
Overall, the feature importance results reinforce the findings from the exploratory analysis, demonstrating that diversion risk is driven by a combination of historical behaviour, destination-specific characteristics, and temporal patterns, rather than any single factor in isolation. This validates the inclusion of engineered historical features and interaction features in the model, as they capture important non-linear and context-dependent relationships that are not fully represented by raw variables alone.

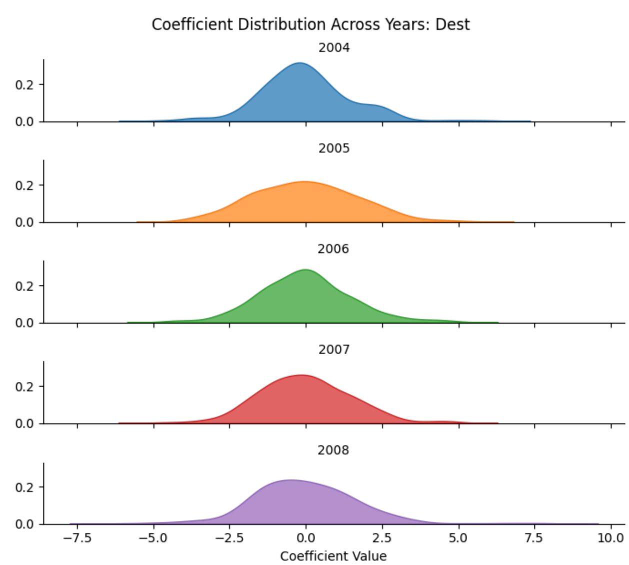
Coefficients Of Categorical Features Across Years

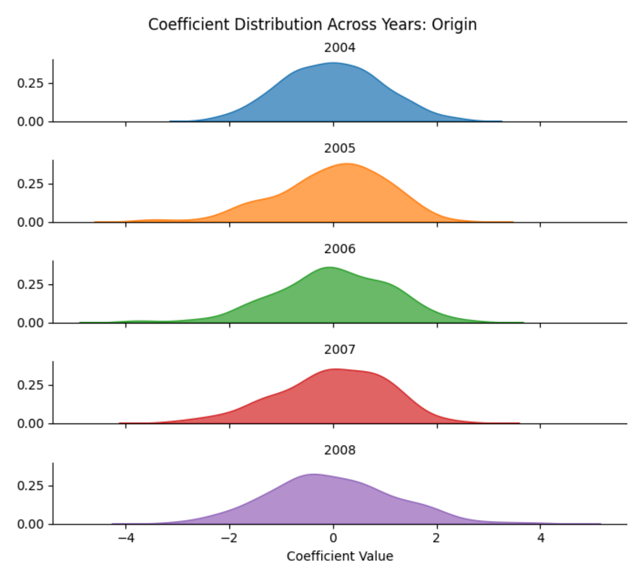


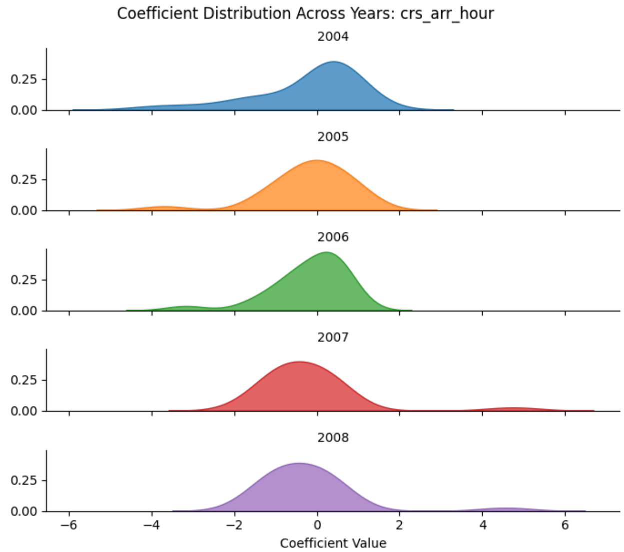


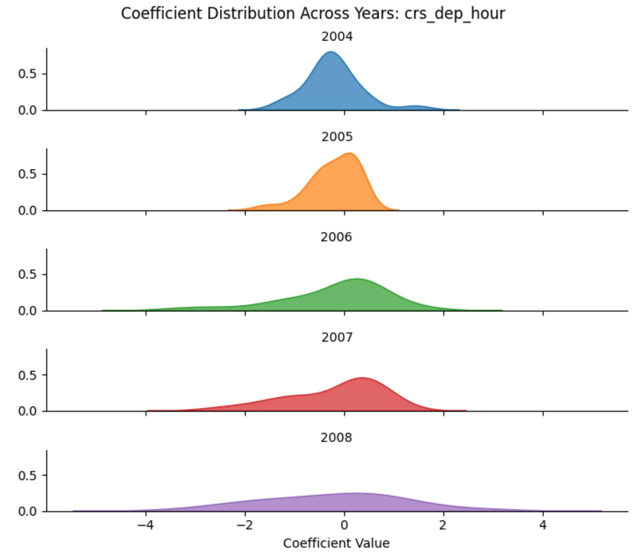


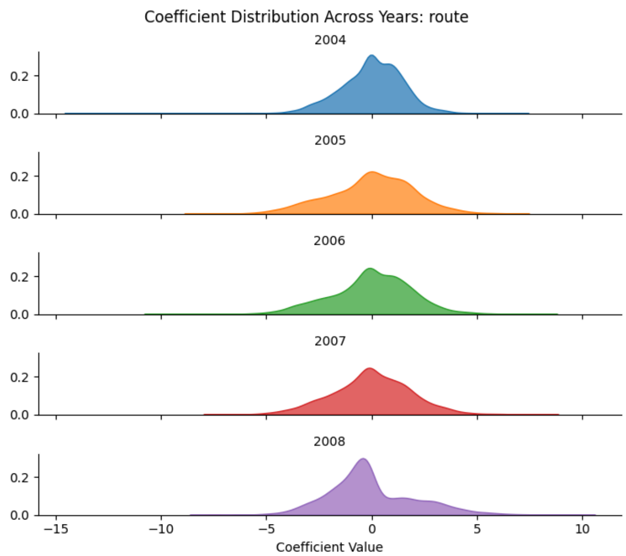












The KDE plots illustrate the distribution of logistic regression coefficients for the encoded categories within each feature group across different years. Rather than focusing on individual categories, the plots provide an overall view of the spread, variability, and relative magnitude of the coefficients associated with each feature group. Wider distributions indicate greater variation in the effects of individual categories, while distributions concentrated near zero suggest weaker or more uniform category-level effects. Overall, destination-related and route-related feature groups exhibit noticeably wider coefficient distributions compared to broader temporal features such as month. This suggests that specific destinations, routes, and airport–time combinations contribute more strongly to diversion risk than general seasonal patterns.

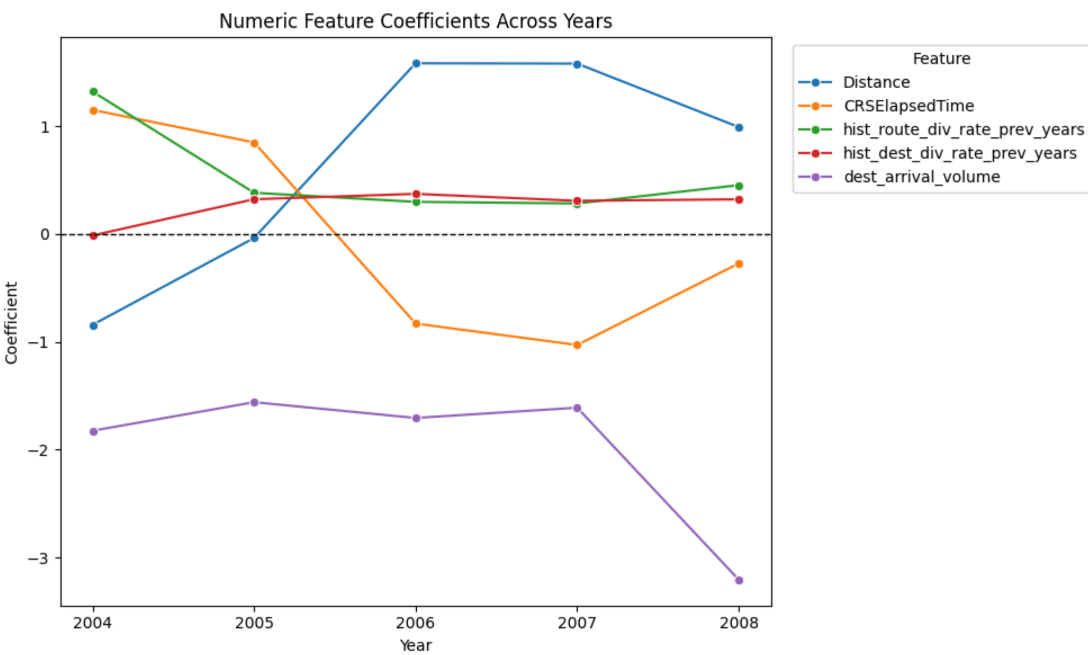
The route coefficients consistently display relatively broad distributions across all years, with both positive and negative tails extending further from zero compared to many other feature groups. This indicates substantial variability in the diversion risk associated with different flight routes, reinforcing the earlier exploratory analysis which suggested that diversion behaviour is highly route-dependent. Similarly, the destination airport (Dest) coefficients also exhibit wide spreads, suggesting that certain destination airports are systematically associated with higher or lower diversion likelihoods.

The interaction-based temporal features, particularly destination–arrival hour (Dest_arr_hr) and origin–departure hour (Origin_dep_hr), show coefficient distributions with extended positive and negative tails across multiple years. Compared to the standalone scheduled hour features, these interaction features demonstrate greater variability, indicating that combining airport and time information captures more granular operational patterns related to diversion risk. This supports the inclusion of the engineered interaction features in the model.

In contrast, the standalone temporal features, namely scheduled arrival hour (crs_arr_hour) and scheduled departure hour (crs_dep_hour), display comparatively narrower and more centrally concentrated distributions. This suggests that while time-of-day effects contribute to diversion prediction, their effects are generally more moderate when considered independently of airport-specific context. The month coefficients exhibit the narrowest and most concentrated distributions among the feature groups, particularly in 2006–2008, where the distributions are clustered closely around slightly negative values. This indicates that seasonal effects are relatively weak predictors of diversion risk compared to operational and location-specific variables. The carrier coefficient distributions remain relatively concentrated around zero in most years, suggesting that differences between airlines contribute less strongly to diversion prediction once operational, temporal, and route-related variables are taken into account. However, the 2007 distribution exhibits a pronounced negative tail, indicating that certain carrier categories may have unstable or strongly negative effects in that year, consistent with the negative permutation importance observed previously.

Overall, the coefficient distributions reinforce the findings from the exploratory analysis and feature importance results. Features associated with routes, destinations, and airport–time interactions exhibit greater variability and stronger category-level effects, indicating that diversion behaviour is driven primarily by operational and contextual factors rather than broad seasonal trends alone.

Coefficients of Numerical Features Across Years



The line plot illustrates the yearly coefficients of the numerical features in the logistic regression models, where positive coefficients indicate an increased likelihood of diversion and negative coefficients indicate a reduced likelihood. The variation in coefficient magnitude and direction across years suggests that the influence of operational factors on diversion risk changes over time rather than remaining fixed.

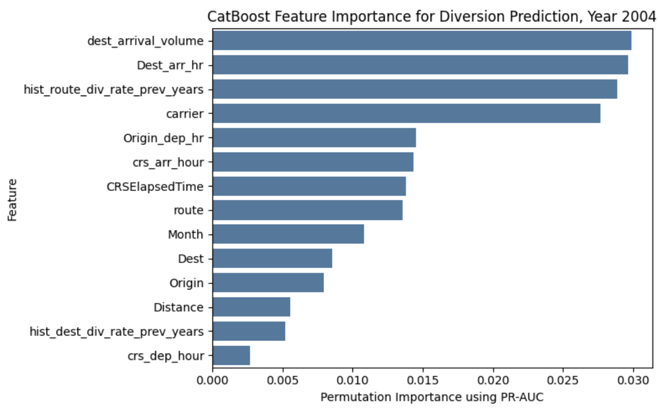
Among the features, Distance exhibits the largest shift, changing from a negative coefficient in 2004 to strongly positive coefficients from 2006 onwards, peaking at approximately 1.5–1.6 in 2006 and 2007. This suggests that longer flight distances became increasingly associated with higher diversion risk in later years. In contrast, CRSElapsedTime displays an opposite trend, transitioning from positive coefficients in 2004–2005 to negative coefficients from 2006 onwards, indicating that longer scheduled flight durations may have reduced diversion likelihood after accounting for other operational variables. The engineered historical features, particularly historical route diversion rate, remain consistently positive across all years, reinforcing the importance of past diversion behaviour in predicting future diversions. The coefficients for historical route diversion rate are generally larger than those for historical destination diversion rate, suggesting that route-level historical patterns provide stronger predictive signals than destination-level history alone. Meanwhile, destination arrival volume remains consistently negative across all years and becomes substantially more negative in 2008. This suggests that airports with larger arrival volumes may possess greater operational capacity or recovery mechanisms that reduce diversion likelihood despite handling higher traffic levels. Overall, the results indicate that both operational and historical features contribute meaningfully to diversion prediction, although their effects vary across years.

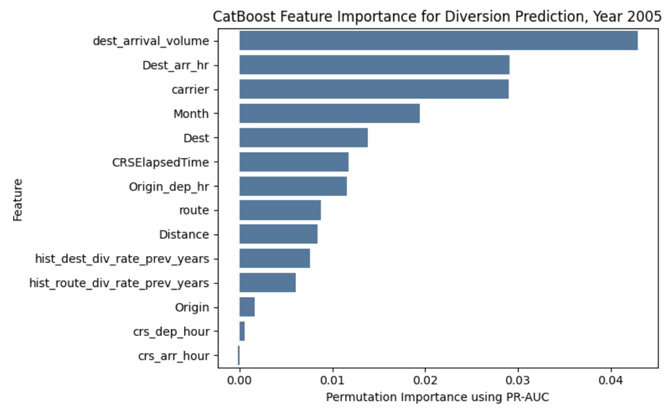
CatBoost Model

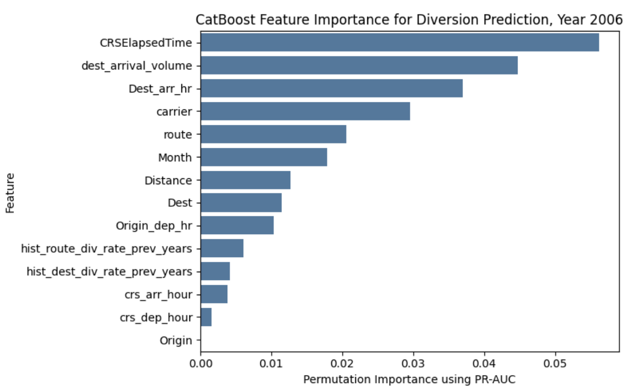
The Logistic Regression model is not the best model for predicting diversions since the model assumes a primarily linear relationship between the features and the log-odds of the target outcome. As seen from the earlier exploratory analysis of the features, it suggests that diversion risk is influenced by complex, non-linear, and interaction-based relationships involving routes, destinations, airports, and time-related variables. Therefore, although logistic regression provides a useful and interpretable baseline model, it may not be the most appropriate approach for predicting flight diversions in this case.

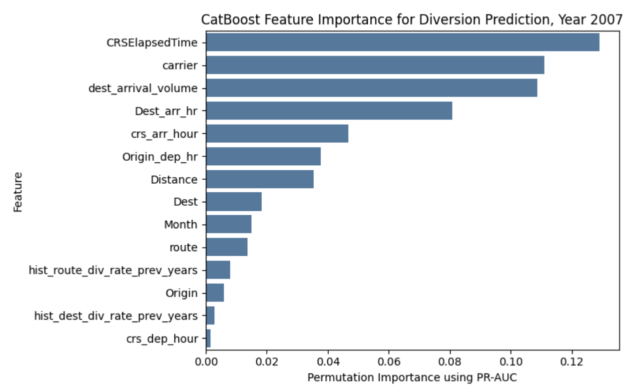
Instead, we choose to use a Catboost model with the same set of features as a more suitable alternative model due to its ability to naturally model non-linear relationships and complex feature interactions without extensive manual feature engineering. CatBoost is able to handle high-cardinality categorical variables effectively, making it particularly well-suited for features such as routes, airports, and carrier categories. Its gradient boosting framework also improves predictive performance on imbalanced classification problems by learning more flexible decision boundaries compared to logistic regression.

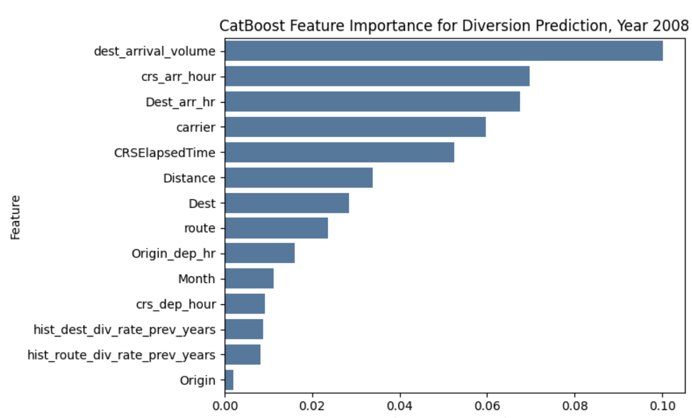
As such, we plot the permutation importance of the CatBoost model for each year as shown below to illustrate how the relative importance of features changes under the CatBoost model that is better able to capture non-linear relationships and feature interactions.











Compared to the logistic regression model, the CatBoost model demonstrates a stronger ability to capture complex and non-linear relationships influencing flight diversions. Under logistic regression, permutation importance values were generally concentrated below approximately 0.02, with the model relying heavily on a smaller group of dominant linear predictors, particularly hist_route_div_rate_prev_years, which recorded the highest importance of approximately 0.012 in 2004. Most remaining variables exhibited relatively small contributions ranging between approximately 0.001 and 0.006.

In contrast, CatBoost distributes importance more broadly across operational, temporal, and interaction-based variables, suggesting that diversion behaviour depends on more complex and context-dependent relationships. One of the clearest examples is CRSElapsedTime, whose importance increases from approximately 0.005 under logistic regression to 0.056 in 2006 and 0.129 in 2007 under CatBoost, where it becomes the strongest predictor. Similarly, the interaction feature Dest_arr_hr increases from approximately 0.005–0.019 under logistic regression to approximately 0.030 in 2004, 0.081 in 2007, and 0.067 in 2008 under CatBoost, indicating that airport–time interactions exert a stronger localized influence on diversion risk than simple linear effects alone.

Operational variables such as carrier and dest_arrival_volume also become substantially more influential under CatBoost. For example, carrier, which exhibited weak or negative importance under logistic regression in some years, increases to approximately 0.111 in 2007 under CatBoost, while dest_arrival_volume increases from approximately 0.006–0.021 under logistic regression to approximately 0.109 in 2007 and 0.100 in 2008. Temporal variables such as Month and crs_arr_hour similarly become more important, suggesting that seasonality and time-of-day effects interact non-linearly with broader operational conditions.

Overall, the broader distribution of feature importance under CatBoost indicates that diversion behaviour is influenced by interconnected operational, temporal, and airport-specific factors rather than isolated linear effects alone. This suggests that CatBoost is better able to identify complex feature interactions and non-linear relationships that may not be adequately captured by logistic regression.

Predictive Performance Comparison

	Year	Base Rate	LR Recall	CB Recall	LR Prec	CB Prec	LR PR-AUC	LR PR-AUC Lift	CB PR-AUC	CB PR-AUC Lift
0	2004	0.0020	0.5741	0.6423	0.0043	0.0049	0.0148	7.5x	0.0486	24.7x
1	2005	0.0020	0.5665	0.6692	0.0045	0.0056	0.0155	7.7x	0.0561	28.0x
2	2006	0.0023	0.6030	0.6672	0.0052	0.0066	0.0131	5.7x	0.0675	29.2x
3	2007	0.0024	0.6211	0.6917	0.0059	0.0083	0.0255	10.8x	0.1625	68.9x
4	2008	0.0024	0.4534	0.6498	0.0068	0.0091	0.0254	10.4x	0.1323	54.4x

Finally, we analyse the predictive performance of the CatBoost (CB) model and compare it to that of the Logistic Regression (LR) model to show how much better it is at predicting diversions.

The predictive performance results indicate that the CatBoost model consistently outperforms the logistic regression model in identifying diverted flights across all years. Although both models were trained on a highly imbalanced dataset with extremely low base diversion rates of approximately 0.20%–0.24%, CatBoost achieves noticeably stronger recall, precision, and PR-AUC values, suggesting that it is better able to detect rare diversion events.

In terms of recall, CatBoost consistently identifies a larger proportion of diverted flights compared to logistic regression. For example, recall improves from 0.574 to 0.642 in 2004, from 0.567 to 0.669 in 2005, from 0.603 to 0.667 in 2006, from 0.621 to 0.692 in 2007, and from 0.453 to 0.650 in 2008. This indicates that CatBoost is more effective at detecting actual diversion cases, particularly in later years where the improvement becomes more substantial.

Precision also improves under CatBoost across all years, although the values remain relatively low due to the rarity of diversion events. For instance, precision increases from 0.0043 to 0.0049 in 2004, from 0.0059 to 0.0083 in 2007, and from 0.0068 to 0.0091 in 2008. While these precision values may appear small in absolute terms, they represent meaningful improvements relative to the extremely low base diversion rates.

The improvement is most clearly reflected in the PR-AUC values. Logistic regression achieves PR-AUC values ranging from approximately 0.013 to 0.026, while CatBoost substantially increases these values to approximately 0.049 in 2004, 0.056 in 2005, 0.068 in 2006, 0.163 in 2007, and 0.132 in 2008. Since PR-AUC is particularly informative for highly imbalanced classification problems, these improvements suggest that CatBoost is considerably better at distinguishing diverted flights from non-diverted flights.

The model performance is also compared against the base diversion rates to contextualise the PR-AUC values relative to the actual prevalence of diverted flights in each year. Since diversion events are extremely rare, absolute PR-AUC values may appear numerically small despite representing meaningful predictive capability. By comparing the PR-AUC values against the observed yearly diversion rates, the lift metric indicates how much more effectively the models are able to identify diverted flights relative to the natural occurrence of diversions within the dataset. This provides a more interpretable measure of practical model performance under severe class imbalance, where even modest increases in PR-AUC can correspond to substantial improvements in identifying rare diversion events.

Comparing the PR-AUC values against the base diversion rates further highlights the effectiveness of the models. Logistic regression achieves PR-AUC lifts of approximately 5.7 times to 10.8 times above the base diversion rate, whereas CatBoost achieves substantially larger lifts ranging from approximately 24.7 times to 68.9 times. The strongest improvement occurs in 2007, where CatBoost achieves a PR-AUC of 0.1625 compared to the base diversion rate of 0.24%, corresponding to a performance lift of approximately 68.9 times above random baseline performance.

Overall, the results suggest that CatBoost is substantially more effective than logistic regression for this diversion prediction task. The stronger recall, higher PR-AUC values, and larger improvements above baseline indicate that CatBoost is better able to identify the complex and non-linear operational relationships associated with flight diversions despite the severe class imbalance present in the dataset.